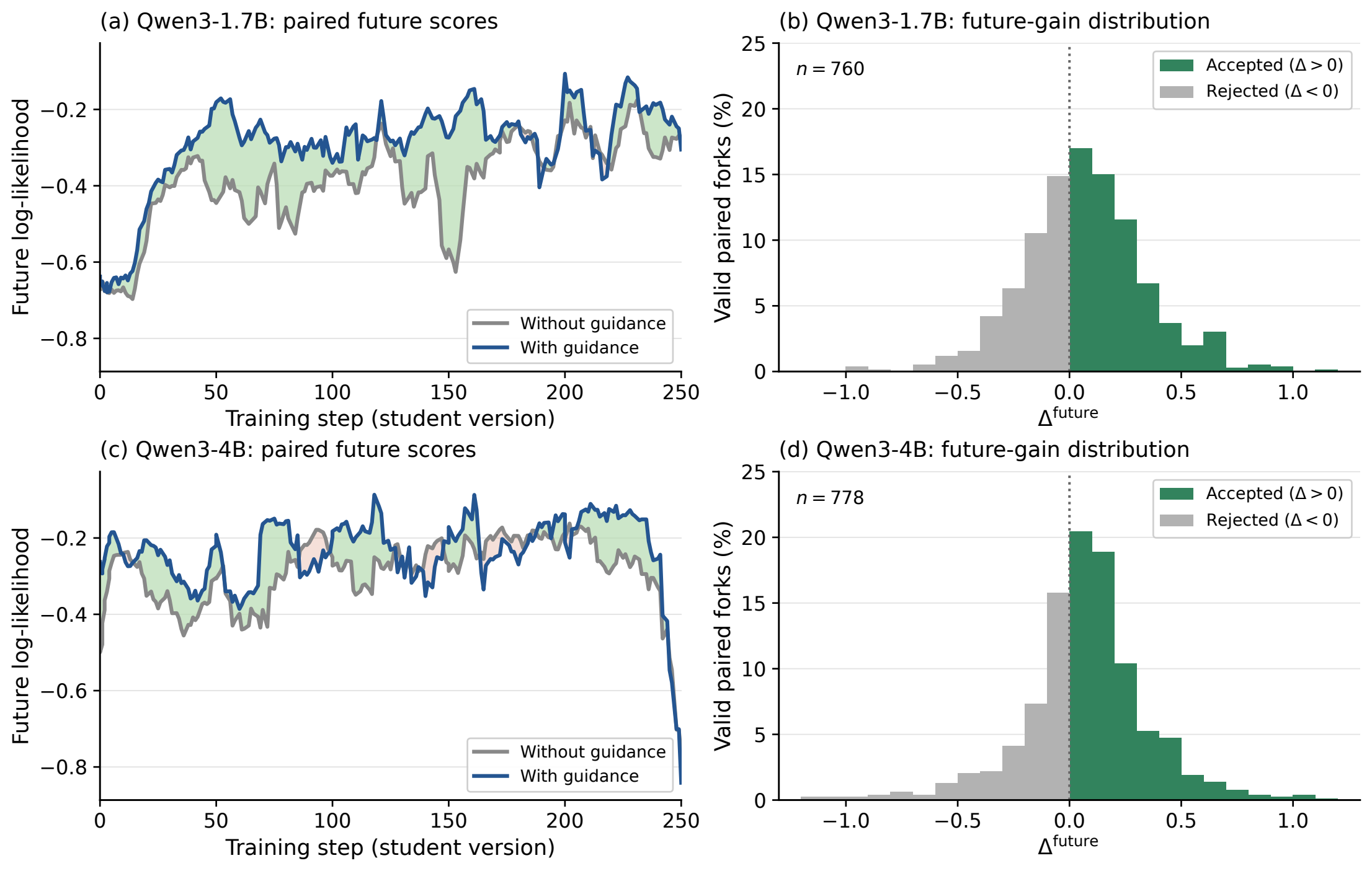
(a) Qwen3-1.7B: paired future scores
Future log-likelihood
−0.2
−0.4
−0.6
−0.8
0
50
100
150
200
250
Training step (student version)
Without guidance
With guidance
(b) Qwen3-1.7B: future-gain distribution
n = 760
Accepted (Δ > 0)
Rejected (Δ < 0)
Valid paired forks (%)
25
20
15
10
5
0
−1.0
−0.5
0.0
0.5
1.0
Δfuture
(c) Qwen3-4B: paired future scores
Future log-likelihood
−0.2
−0.4
−0.6
−0.8
0
50
100
150
200
250
Training step (student version)
Without guidance
With guidance
(d) Qwen3-4B: future-gain distribution
n = 778
Accepted (Δ > 0)
Rejected (Δ < 0)
Valid paired forks (%)
25
20
15
10
5
0
−1.0
−0.5
0.0
0.5
1.0
Δfuture